

Freight Rate Prediction

Model evaluation report | October 2026

Executive summary

CatBoost was selected as the leading candidate on a chronological holdout (MAE USD 131.57). It was then refit on all 48,000 labeled loads for final inference.

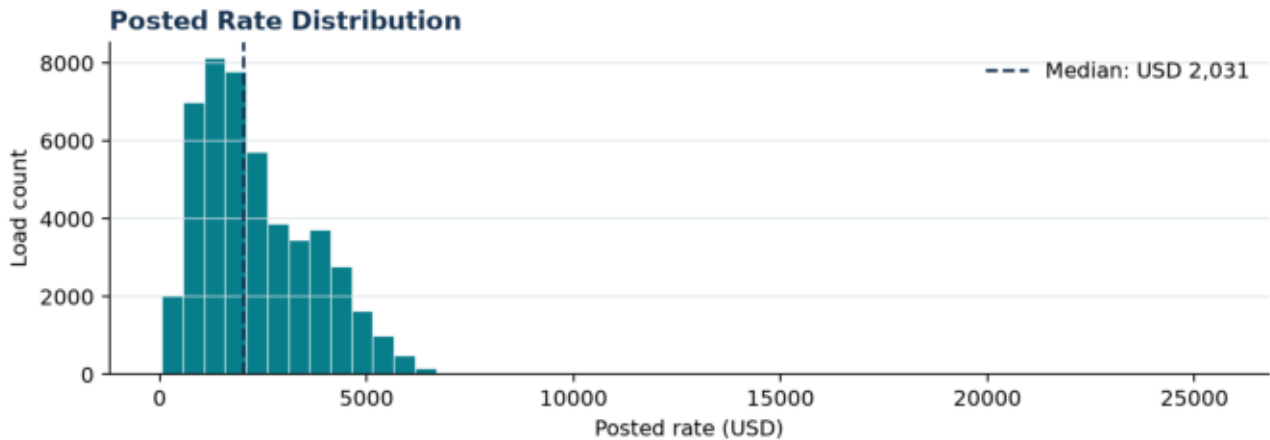
1. Problem Overview

Predict posted freight rates for loads using lane, distance, equipment, weight, location coordinates, date, market index, and quote signal. The objective is to estimate rates for future loads.

2. Data Exploration

- Development data: 48,000 labeled loads and 14 source columns. Target variable: posted_rate.
- Predictors include pickup and delivery cities and coordinates, distance, equipment, weight, date, market_index, and quote_signal. Labeled dates run from 2025-01-01 through 2025-10-31.
- Posted rates exhibit right-skewness (skewness 1.90) with a small number of high-value loads. Median USD 2,030.76; 99th percentile USD 5,972.83; maximum USD 25,533.

Target Distribution



GitHub Repository: <https://github.com/hamzansour43/spotter-freight-rate-assessment>

Data Quality and Feature Engineering

Quality checks and transformations applied before modeling

3. Data Quality Findings

- Missing values: weight in 300 rows (0.625%) and market_index in 374 rows (0.779%). Linear Regression and Random Forest use numeric median imputation learned from training data; CatBoost handles numeric missing values natively.
- 292 negative weights (minimum -47,500). In model copies, negative weight and derived weight_per_mile are treated as missing; original data is unchanged. Investigate source records before production.
- No full duplicate rows, non-positive distances, or non-positive posted rates were found. Equipment categories: Dry Van, Reefer, and Flatbed.

4. Feature Engineering

- month, day_of_week (Monday=0), ISO week, and quarter are extracted from date to represent calendar effects.
- distance_squared adds a nonlinear distance term; weight_per_mile is weight divided by positive distance. Invalid negative-weight-derived values are marked missing in model copies.
- lane combines pickup and delivery city names (pickup_delivery). pickup, delivery, equipment, and lane are categorical; missing category values use __MISSING__.
- Original numeric features retained: four location coordinates, distance, weight, market_index, and quote_signal. load_id is excluded; raw date is represented through date features.

CatBoost uses native categorical handling; Linear Regression and Random Forest use one-hot encoding.

Validation Strategy and Models

Time-aware evaluation design and candidate model configurations

5. Validation Strategy

A chronological holdout approximates forecasting into the future and avoids training on dates later than those evaluated. All loads from each date remain in the same partition.

- The 304 unique labeled dates were split at 2025-09-01. Training: 38,477 loads dated 2025-01-01 to 2025-08-31. Holdout: 9,523 loads dated 2025-09-01 to 2025-10-31.
- Preprocessing was learned on the training partition only. The separate 12,000-row validation set (2025-11-01 to 2025-12-31) has no target labels and was reserved for prediction, not model selection.

6. Models Evaluated

- Linear Regression: numeric median imputation and one-hot encoding; simple baseline.
- Random Forest: 300 trees, max_features=0.5, min_samples_leaf=2, random seed 42; numeric median imputation and one-hot encoding.
- CatBoost: 800 iterations, depth 8, learning rate 0.05, RMSE objective, random seed 42; native categorical handling for pickup, delivery, equipment, and lane.

All models use the same engineered features and Sep-Oct 2025 holdout. The unlabeled final validation data was not used to calculate these metrics.

Results and Final Model Selection

Holdout performance | Sep-Oct 2025

7. Results

Lower is better. MAE is the primary selection metric; RMSE gives greater weight to large errors.

Model	MAE	RMSE	MAPE
Linear Regression	USD 187.14	USD 649.07	8.37%
Random Forest	USD 139.19	USD 655.53	6.07%
CatBoost	USD 131.57	USD 642.17	6.12%

8. Final Model Selection

CatBoost led on the primary metric MAE (USD 131.57) and on RMSE (USD 642.17). Its MAE was USD 55.57 (29.7%) lower than Linear Regression and USD 7.62 (5.5%) lower than Random Forest. Random Forest had a marginally lower MAPE (6.07% vs. 6.12%). CatBoost was selected based on the prespecified MAE priority and its lowest RMSE.

After comparison, CatBoost was refit on all 48,000 labeled loads for final inference.

Business conclusion

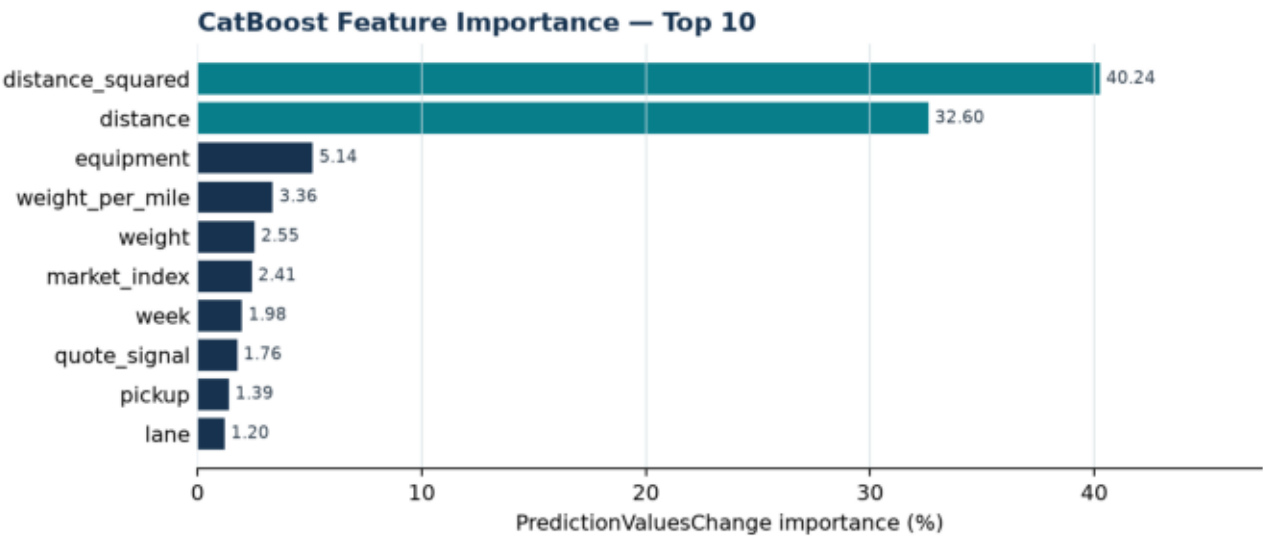
The final CatBoost model achieved approximately 6.1% holdout MAPE, suggesting useful predictive accuracy for freight-rate estimation under similar market conditions.

Feature Importance and Limitations

Interpretation of the fitted full-data CatBoost model

CatBoost Feature Importance

PredictionValuesChange importance is relative model contribution, not a causal effect. Distance and distance_squared dominate (72.85% combined); lane and pickup contribute less in this fitted model.



Limitations

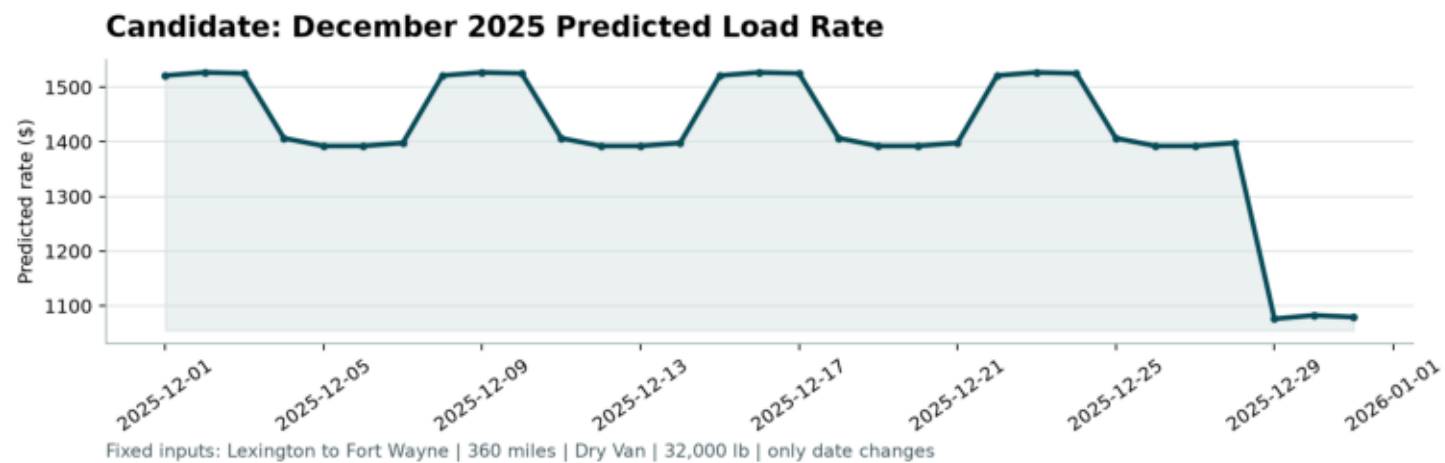
The model was evaluated on a single chronological holdout period. Additional rolling-window validation could provide a more robust estimate of future production performance.

The tree-model hyperparameters are initial settings, not extensively tuned. The 292 negative-weight records also require source review; they were treated as missing only in model input copies.

December Prediction Chart

9. December Prediction Chart

The scorer-validated daily predictions cover December 1-31, 2025. Pickup, delivery, distance, equipment, and weight are fixed; only date changes by day.



Interpretation: predicted rates remain within USD 1,392-1,527 for December 1-28, with recurring fluctuations. They drop to approximately USD 1,076-1,083 on December 29-31. With lane and shipment details fixed, the model attributes these changes to learned date/calendar patterns; this is a prediction, not observed market movement.

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